

1)

ASCII to binary

alpha = 0110000101101100011100000110100001100001

bravo = 0110001001110010011000010111011001101111

delta = 0110010001100101011011000111010001100001

gamma = 0110011101100001011011010110110101100001

c1 = 1111100101111001110011000001011110000110

c2 = 1111101001100111110111010000100110001000

c1 xor alpha = 1001100000010101101111000111111111100111

c2 xor bravo = 1001100000010101101111000111111111100111

c1 xor delta = 1111110001110000110100000000101110000110

c2 xor gamma = 1001110100000110101100000110010011101001

To verify the key from the plain text and the cipher text one just has to xor the cipher text with the plain text and that should result in the key. This is because the cipher text is created by xoring the key with the plain text, doing the xor again reverses that operation. Once I have the key from each encryption then it is simply making sure the key from c1 xor Alpha matches c2 xor Bravo or that c1 xor delta matches c2 xor gamma.

In this case the resulting key from c1 xor alpha and c2 xor bravo match while the latter pair does not. The key for this encryption is 1001100000010101101111000111111111100111

2)

The underlying flaw with this argument is that EAVESDROP is a randomized algorithm. That means that if you call EAVESDROP several times even on the same input you are likely to get different outputs. Also that the specific  $m$  chosen has no connection to  $c$ . IE one could choose another arbitrary  $m'$  and  $EAVESDROP(m')$  will result in the exact same randomized algorithm, meaning any  $m$  will result in an uniform distribution.

|                                                                                                              |
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| $\text{EAVESDROP}'(m \in \mathbb{Z}_n):$<br>$k \leftarrow \mathbb{Z}_n$<br>$c := (k + m) \% n$<br>return $c$ |
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3) To show that the output of  $\text{EAVESDROP}'(m \in \mathbb{Z}_n)$  has a uniform distribution I will pick a  $n$  value and show all the possible values. If each unique output has the same number of occurrences, then that means the probabilities of each are also the same, IE a uniform distribution.

Fix  $n = 2$

$C = (k + m) \% 2$

| K | m | c |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 1 | 2 |
| 2 | 2 | 1 |
| 0 | 1 | 1 |
| 1 | 2 | 0 |
| 2 | 0 | 2 |
| 0 | 2 | 2 |
| 1 | 0 | 1 |
| 2 | 1 | 0 |

From the above table you can see that when  $n = 2$  there are 3 outputs of 0,1, and 2, meaning that the probability of any one is  $1/3$ . Because each output has the same probability that means the output of this function has a uniform distribution.